

RAMSES

Total Presentation Time: 7 min

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RAMSES Main Routines

HYDRO

- Godunov MUSCL-Hancock HLLC Riemann solver
- 2 kernels: **cube** (Bob Caddy) and **paper** (Troels Haugbolle)
- nsubgrid = 1, 2 or 4

PARTICLES

- Mass deposition
- Kick-Drift-Kick pusher

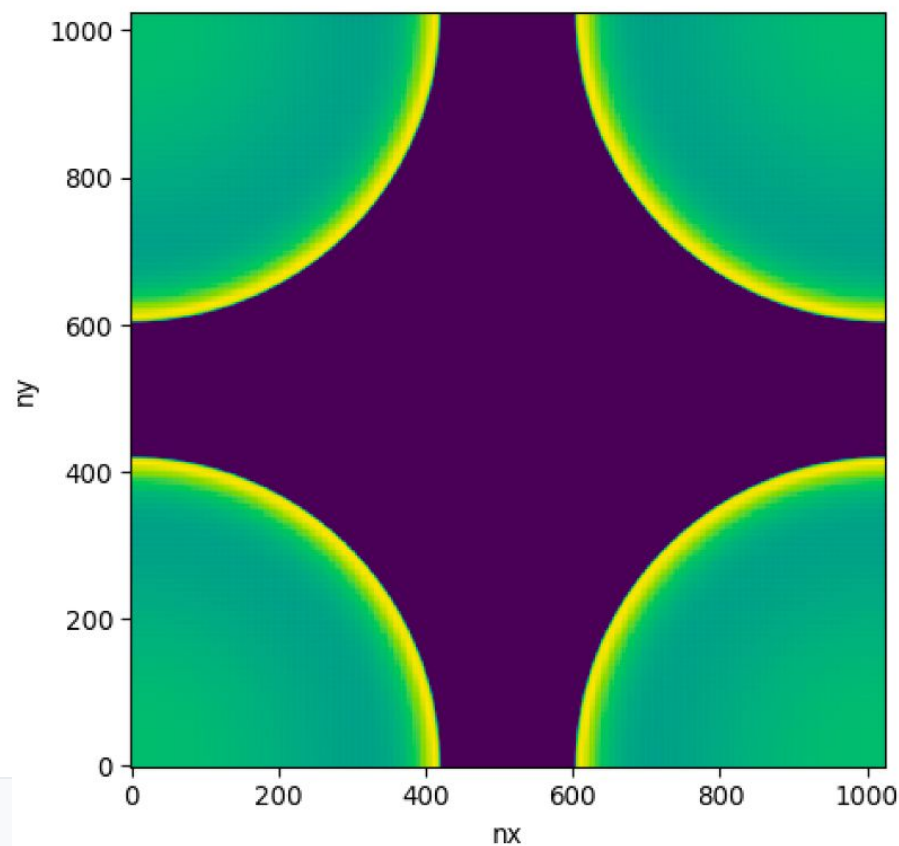
GRAVITY

- Multigrid (MG) Poisson solver (only nsubgrid = 1)
- Hierarchical multi-resolution Gauss-Seidel iteration (V-cycle)

AMR

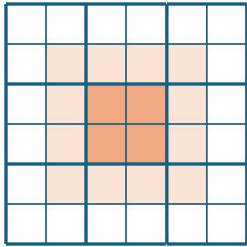
- Refinements / de-refinements
- Ghost zones

HYDRO SOLVER

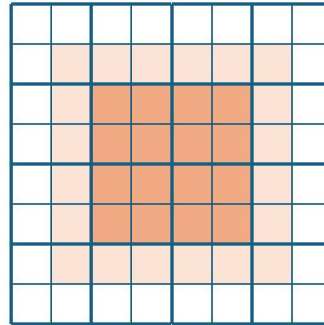


Higher performance with larger stencil

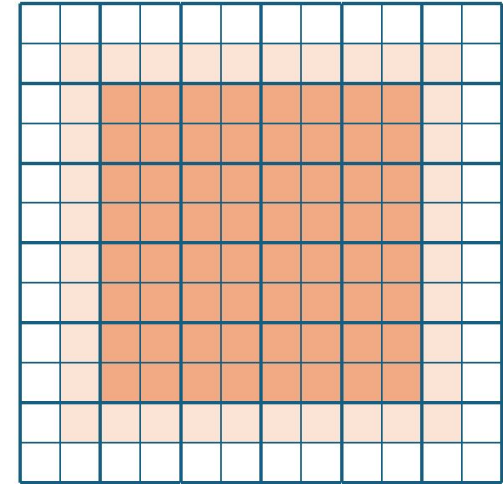
nsubgrid=1
216 cells



nsubgrid=2
512 cells



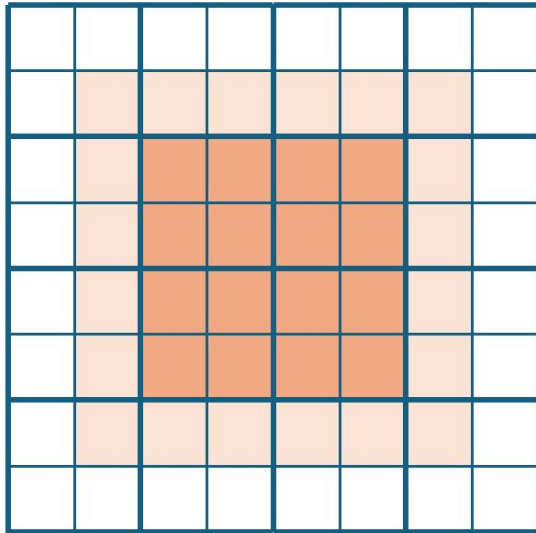
nsubgrid=4
1728 cells



Less shared memory usage with paper kernel

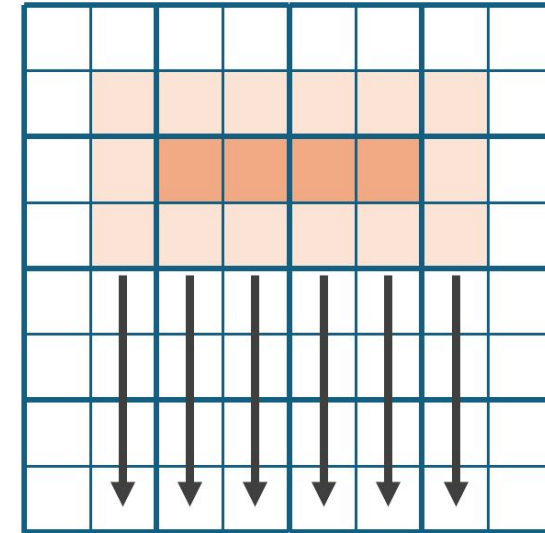
Rock kernel

nsubgrid=2 | 512 cells

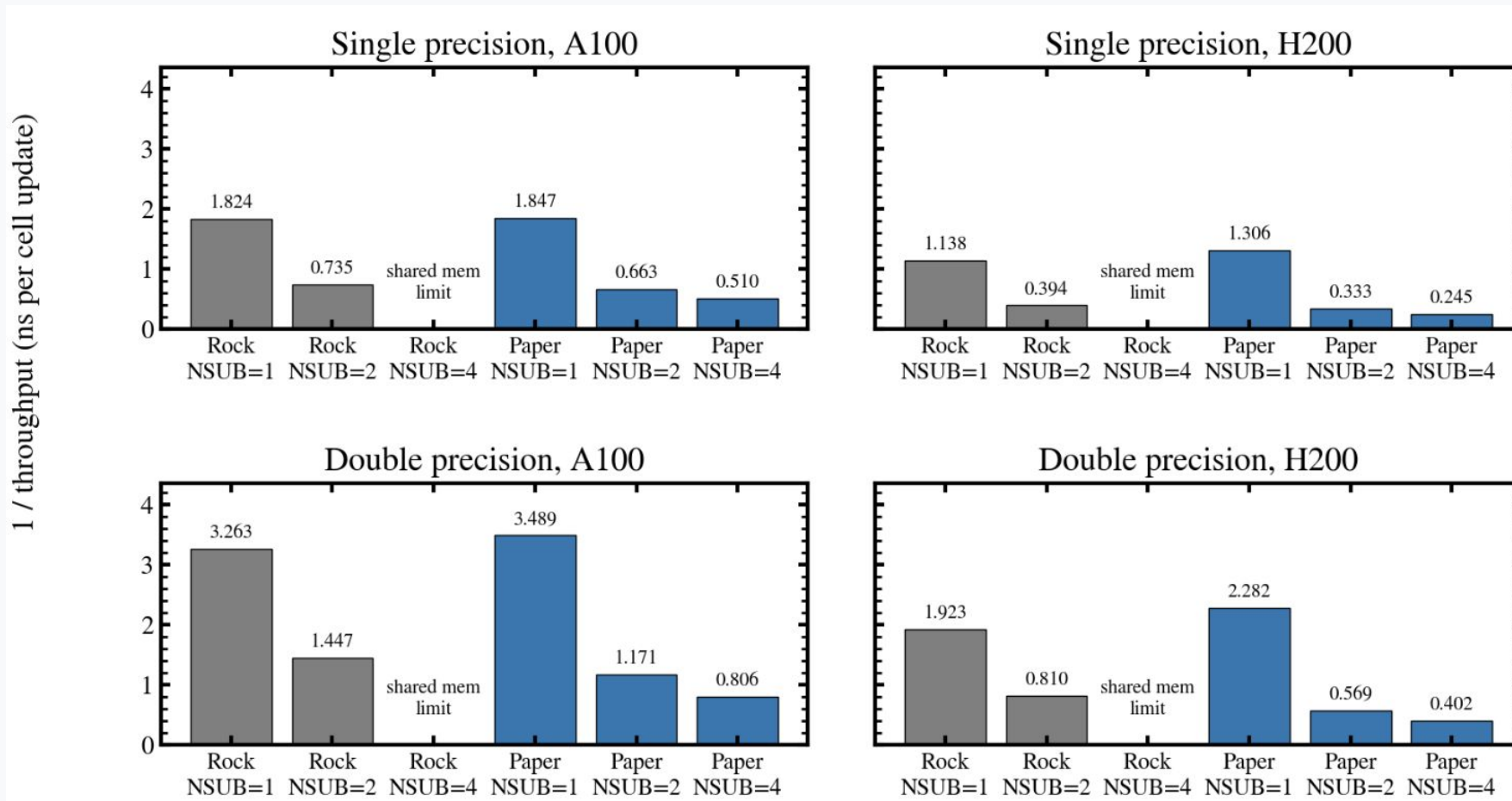


Paper kernel

nsubgrid=2 | 192 cells

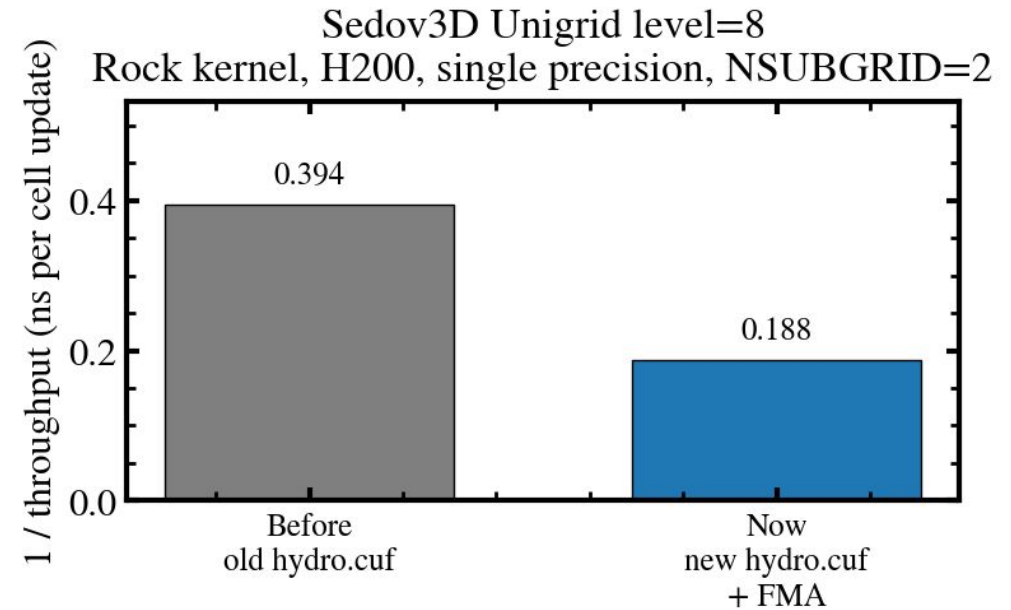


Hydro kernels benchmark (Sedov3d unigrid)



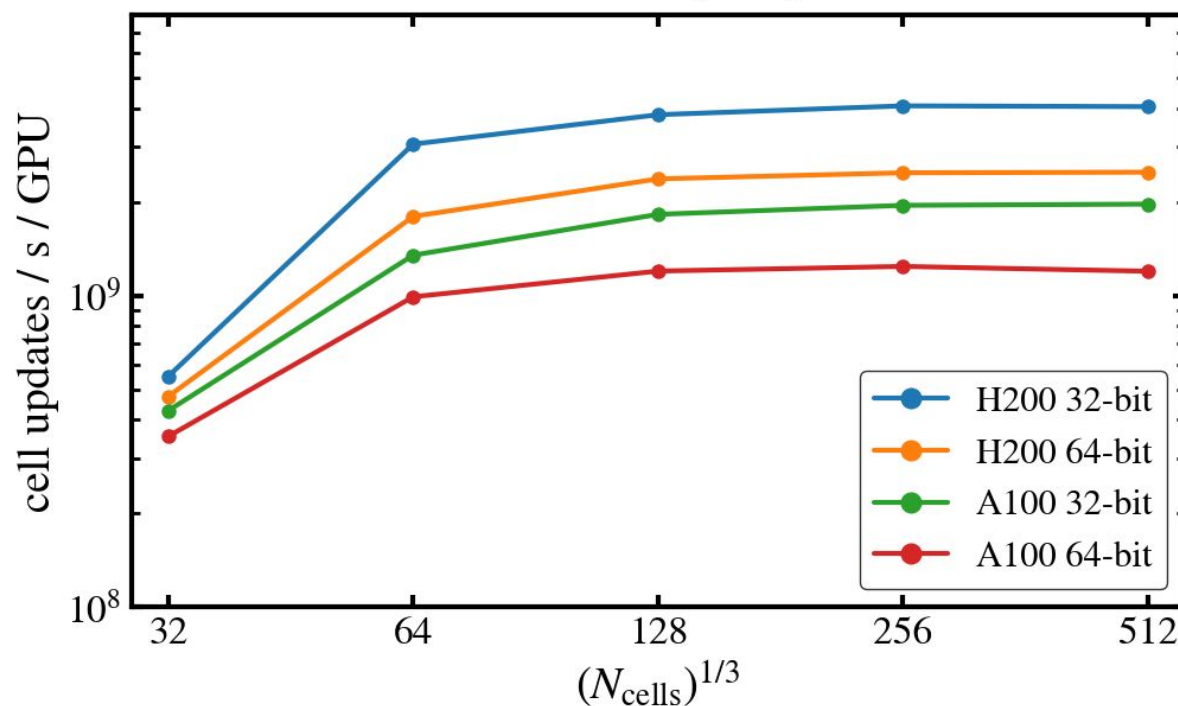
Hydro kernel (rock) additional optimizations

- Matching the precision of float arithmetics
- Minimizing the number of divisions
- `gpu=fastmath`

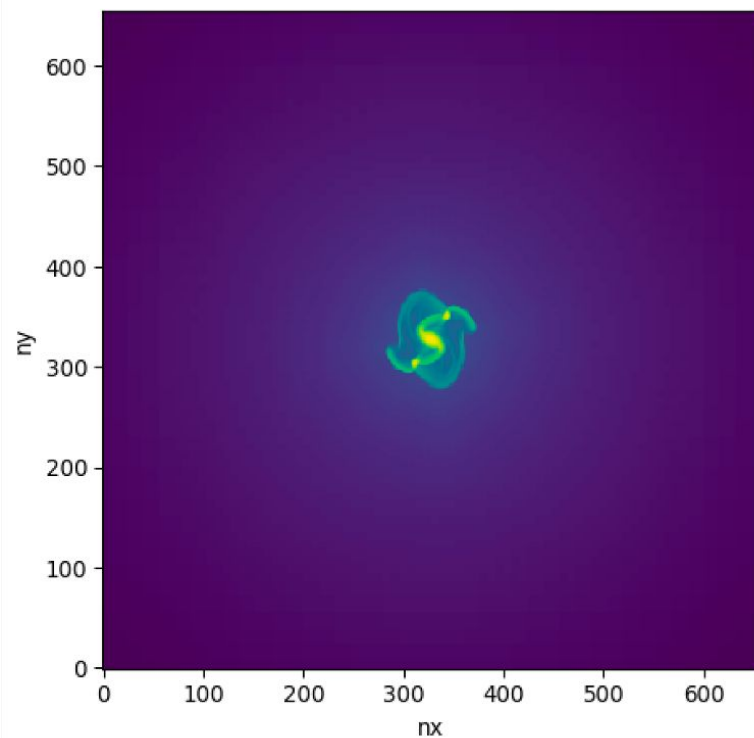


Hydro kernels benchmark (Sedov3d unigrid)

Sedov3D Unigrid: Paper Hydro kernel, NSUBGRID=4
No-I/O coarse-step elapsed time

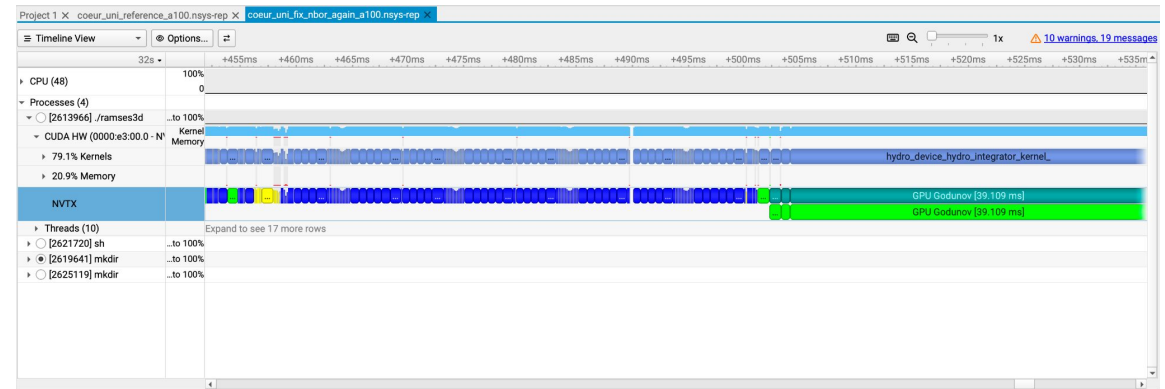
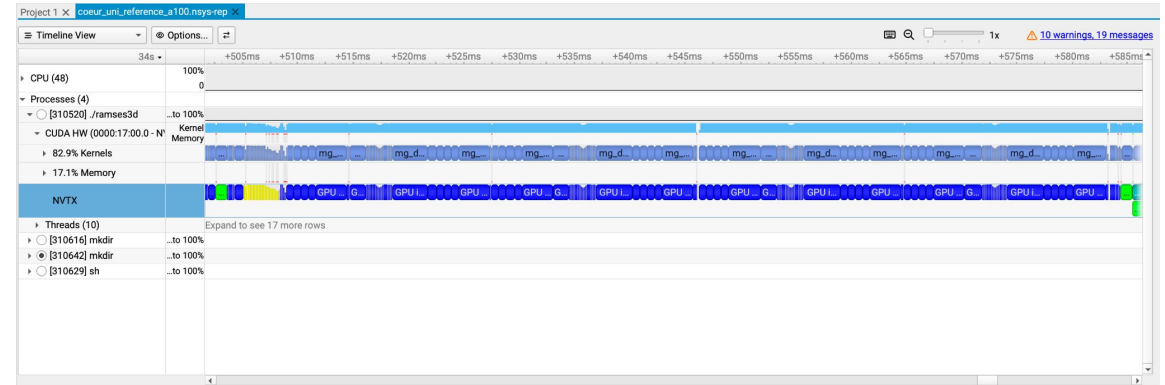


GRAVITY SOLVER



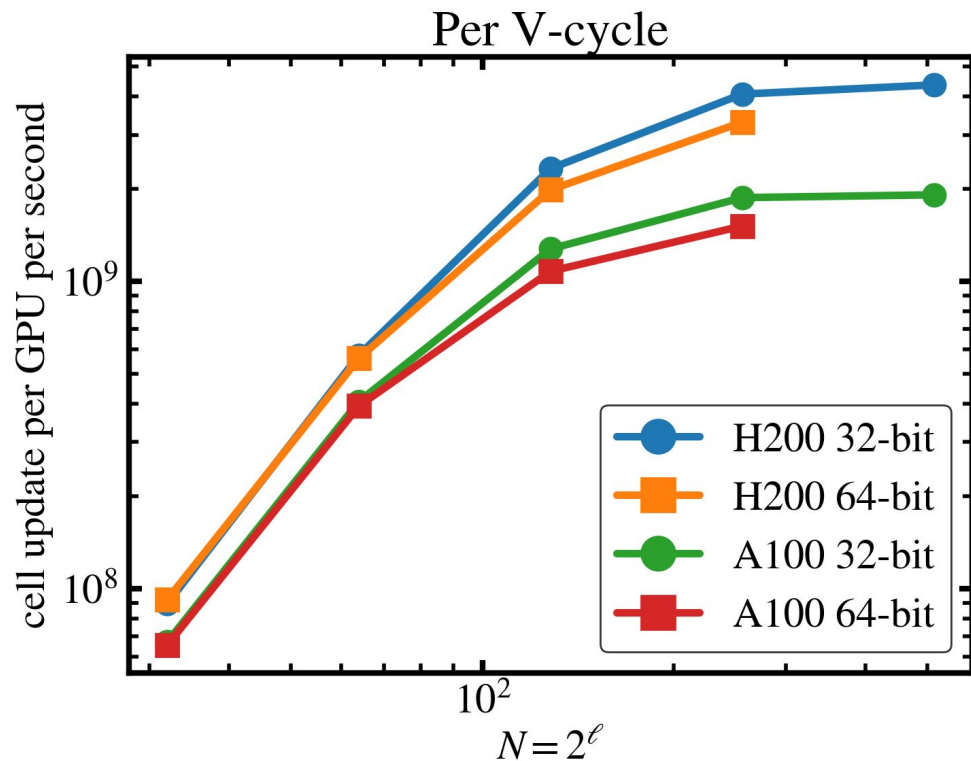
NSight Systems Output and Easy Optimizations

- Minimized D2H and H2D memory copies
- Fused kernels
- Minimized number of main GPU memory access



Progress and Goals

- We have profiled the code with NSight Systems and Compute
 - Identified a few low hanging fruits
 - too many integer operations
- Benchmarked gravity MG solver on A100 (H200) with SP and DP
 - V-cycle: 2 (4) billion cell updates per GPU/s
 - MG hierarchy initialization: $\frac{1}{2}$ V-cycle
- Optimizations
 - We fused several kernels
 - We reduced the number of D2H copies.



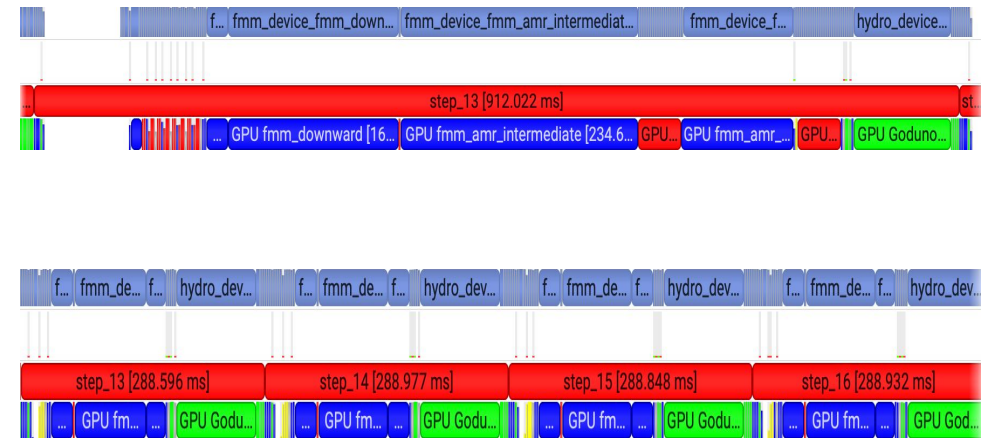
Alternative Gravity Solver: FMM

Algorithm

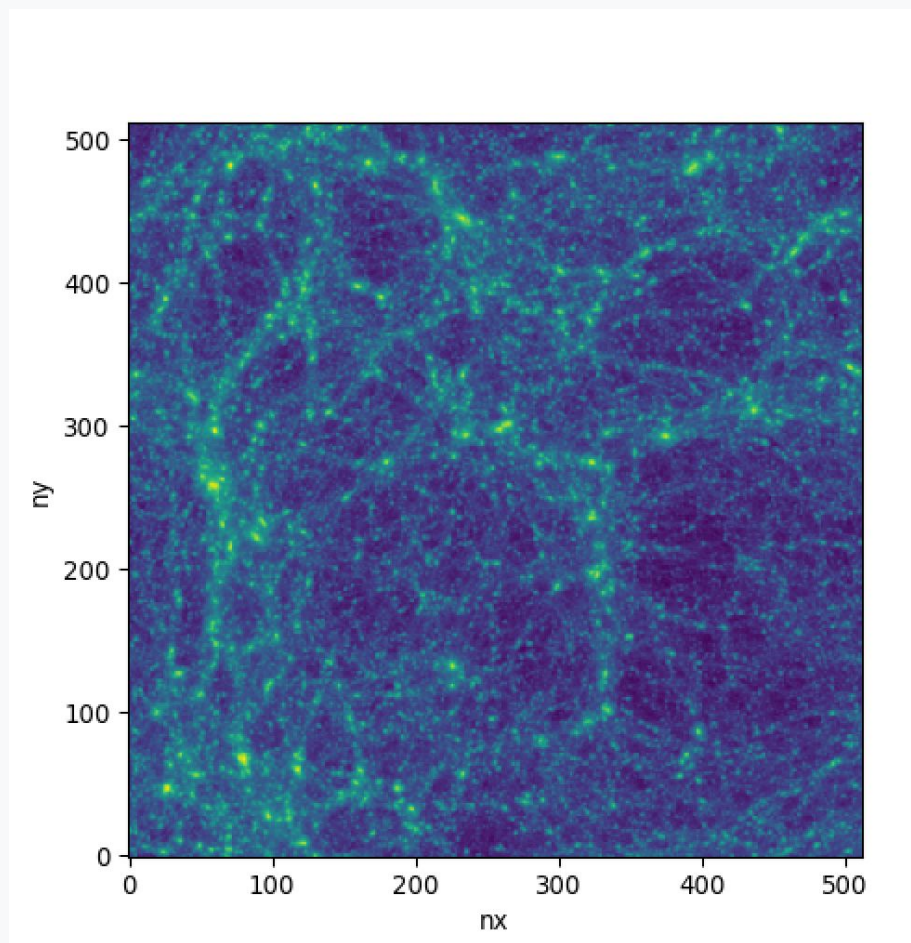
- Downward (**6.21x**)
- Intermediate (**2.74x**)
- Direct (**4.64x**)

Bottlenecks and solutions

- Lower local register pressure / spilling
- Better memory coalescing
- Reduced ALU Ops using precomputation and caching
- Loop removal / arithmetic inlining
- *Reached 6-70% compute, 7-80% memory throughput.*



PARTICLE SOLVER



NSight Systems Reports

Algorithm

- Sort particles
- Cloud-in-cell deposit particles onto grid
- Multigrid to solve poisson's equation (gravity)
- Compute forces from gravitational potential and update positions & velocities

Bottlenecks and solutions

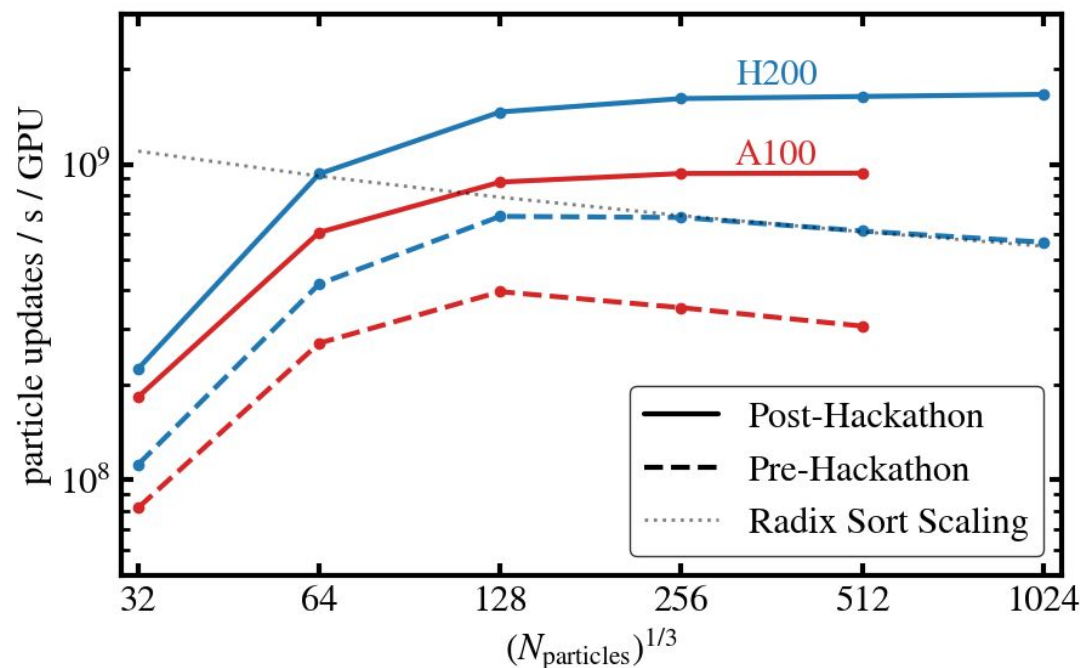
- CIC deposition: we optimized the algorithm
- Radix Sort: we use the CUB library



Performance Analysis: Sort/Mass Deposition

Roofline Analysis

- Performance increases with problem size as it saturates the GPU.
- Once saturated, performance decreases due to **sort algorithm scaling**.
- With all of our optimizations we achieved **~3x speed up**.
- We are approaching 2 billion particle updates per GPU per sec



All together:
Isothermal hydro
+ DM + AMR

